Automotive Engineering 🔧

Brownie Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

Today we make it MOVE 🏁

You designed a car. Now you'll **engineer** one that really rolls — then make it **better**.

What makes a car go? 🏔️

No engine today. We use a **ramp** and **gravity**!

- **Force** = a push or a pull 💪
- **Friction** = grippy slows you down 🐌, slippery lets you roll 💨

Predict first! 🔮

Will a heavier car or a lighter car roll farther?

Everybody guess!

Engineers always predict before they test.

Build your roller 🛠️

Make a cardboard car with: - **Axles** that go straight across - **Wheels** that spin freely

Goal: it rolls straight down the ramp.

Wobbly wheels? That's a **real engineering problem** — let's solve it together!

Test. Change. Test again. 🔁

This is the heart of engineering:

1. Roll your car down the ramp 🏁
2. **Measure** how far it went 📏
3. Write it down 📝
4. Change **ONE** thing
5. Test again — did it help?

The rule: ONE thing at a time

Change **only one thing** between tests.

Change five things and you won't know **which one worked**.

That's called a **fair test**.

What could you change? 🤔

- Add weight ⚖️
- Swap the wheels 🛞
- Straighten the axle 📏

Pick ONE. Predict. Test. Record.

Robot moment 🤖

Our team tests the robot over and over, changing **one thing at a time**, to make it work better.

Same fair-test habit as yours!

Best IMPROVEMENT wins 🏆

Not “whose car is fastest.”

“Who made the biggest improvement?”
Share one change you made and what happened.

You did it! 🎉

Today you: - ✅ Learned that a force like gravity makes a car move - ✅ Built a car with wheels that roll - ✅ Measured how your car did - ✅ Changed one thing, tested again, and knew if it improved

You are an engineer now!

Keep going at home! 

- Make a ramp from a book and test toy cars
- Try a fair test at home: change ONE thing and measure
- Ask your troop leader about the next Automotive badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.